

Midterm Formula sheet
Statistical Computation and Analysis 2026-2

Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \text{ where } P(B) > 0$$

Bayes' Rule

$$P(\theta | y) = \frac{P(y | \theta)P(\theta)}{P(y)}$$

where:

$$P(y) = \sum_{\theta} P(y | \theta)P(\theta)$$

Proportional form:

$$P(\theta | y) \propto P(y | \theta)P(\theta)$$

Common Distributions and Models

Bernoulli

$$y_i \sim \text{Bernoulli}(\theta)$$
$$P(y_i | \theta) = \theta^{y_i}(1 - \theta)^{1-y_i}$$

Where:

$$y_i \in \{0,1\}$$

and

$$\theta = P(y_i = 1)$$

Mean:

$$E[y_i] = \theta$$

Mode:

$$\text{mode}(y_i) = \begin{cases} 1, & \theta > 0.5 \\ 0, & \theta < 0.5 \\ 0 \text{ and } 1, & \theta = 0.5 \end{cases}$$

Variance:

$$\text{Var}(y) = \theta(1 - \theta)$$

Binomial

$$z \sim \text{Binomial}(n, \theta)$$
$$P(z | n, \theta) = \binom{n}{z} \theta^z (1 - \theta)^{n-z}$$

Where:

$$z = \sum_{i=1}^n y_i$$

n = number of trials

θ = probability of success

Mean:

$$E[z] = n\theta$$

Mode:

$$\text{mode}(z) = \lfloor (n+1)\theta \rfloor$$

if $(n+1)\theta$ is an integer, there are two modes: $(n+1)\theta - 1$ and $(n+1)\theta$.

Variance:

$$\text{Var}(z) = n\theta(1-\theta)$$

Poisson

$$y_i \sim \text{Poisson}(\lambda)$$
$$P(y_i | \lambda) = \frac{(\lambda^{y_i} e^{-\lambda})}{y_i!}$$

Where:

$$y_i \in \{0, 1, 2, 3, \dots\}$$

and:

$$\lambda > 0$$

Mean:

$$E[y] = \lambda$$

Mode:

$$\text{mode}(y) = \lfloor \lambda \rfloor$$

if λ is an integer, there are two modes: $\lambda - 1$ and λ .

Variance:

$$\text{Var}(y) = \lambda$$

Beta

$$\theta \sim \text{Beta}(\alpha, \beta)$$
$$p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

Where:

$$0 \leq \theta \leq 1$$

Mean:

$$E[\theta] = \frac{\alpha}{\alpha + \beta}$$

Mode:

$$\text{mode}(\theta) = \frac{(\alpha-1)}{(\alpha+\beta-2)}, \alpha > 1, \beta > 1$$

Variance:

$$\text{Var}(\theta) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Gamma

$$\lambda \sim \text{Gamma}(\alpha, \beta)$$
$$p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

Where β is the rate
and:

$$\lambda > 0, \alpha > 0, \beta > 0$$

Mean:

$$E[\lambda] = \frac{\alpha}{\beta}$$

Mode:

$$\text{mode}(\lambda) = (\alpha-1)/\beta, \alpha > 1$$

Variance:

$$\text{Var}(\lambda) = \frac{\alpha}{\beta^2}$$

MLE

Binomial/Bernoulli

$$\hat{\theta}_{MLE} = \frac{z}{n}$$

or:

$$\hat{\theta}_{MLE} = \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Poisson

$$\hat{\lambda}_{MLE} = \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Useful functions (PyMC and Arviz)

```
pm.Beta("theta", alpha=a, beta=b)
```

```
pm.Gamma("lam", alpha=a, beta=b)
```

```
pm.Bernoulli("y", p=theta, observed=data)
```

```
pm.Binomial("y", n=n, p=theta, observed=data)
```

```
pm.Poisson("y", mu=lam, observed=data)
```

```
pm.sample(draws, chains=4)
```

```
az.plot_trace(idata)
```

```
az.plot_posterior(idata)
```

```
az.summary(idata)
```

```
az.extract(idata)
```